

SECURITY ALARM For Two-Wheelers

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This simple security alarm circuit can help you secure your bike or scooter. It raises an alarm if someone tries to drive away your two-wheeler. You just need to connect the circuit to the vehicle's wheel (preferably front wheel) through a thin flexible wire that can break easily.

Connect the bike's wheel to the circuit with the wire in such a way that when someone tries to steal the bike, the wire would break. When that happens, the speaker in the circuit would generate a machine-gun sound, or some other alarming sound, and alert everyone around.

The prototype assembled on a breadboard for testing in EFY Lab is shown in Fig. 1. Circuit diagram of the security alarm is shown in Fig. 2.

Circuit and working

The alarm circuit is built around machine-gun sound generator IC UM3562 (IC1), transistors BC547 (T1) and SL100 (T2), a 3.1V Zener diode (ZD1), rectifier diode 1N4007 (D1), and a loudspeaker (LS1).

Sound generator IC UM3562, the heart of the circuit, is a low-power CMOS LSI (large-scale integration) chip in an 8-pin DIP package. It includes a voltage-controlled oscillator (VCO), timer generator, tristate selector switch, envelop control, and mixer.

Normally pin 2 of UM3562, along with trigger pin 4, is used as a tri-state selector switch that helps generate three types of sound. When pin 2 is low, a special kind of pulsating sound is generated. When it is connected to 3V, it generates a rifle's

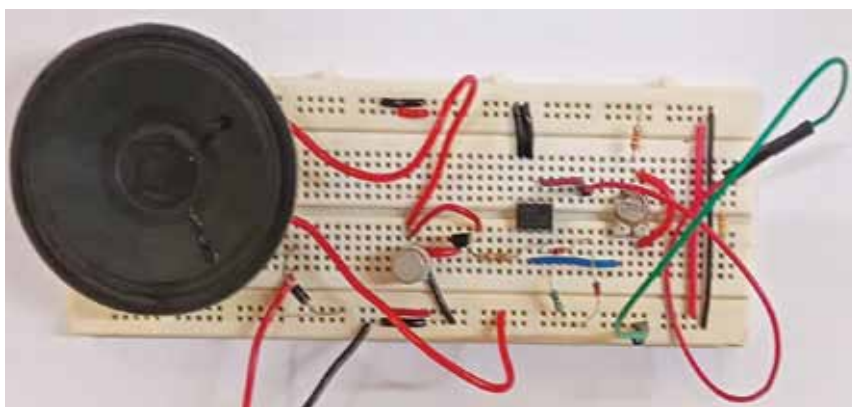


Fig. 1: Prototype tested on breadboard

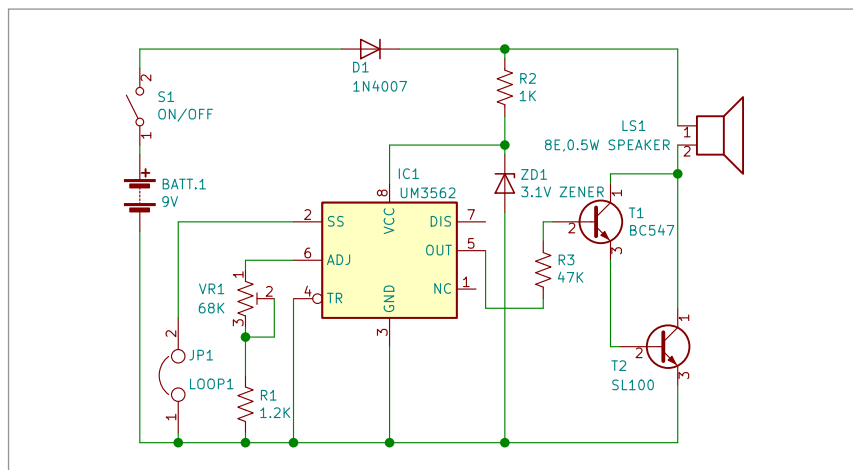


Fig. 2: The circuit diagram

firing sound. When this pin is in floating condition (unconnected), it generates the machine-gun sound.

In this design, pin 2 is kept in low state by making trigger pin 4's state low. When the wire connecting the circuit to vehicle's wheel breaks, machine-gun sound is generated. The machine-gun's shooting rate (sound) can be adjusted through preset VR1.

IC UM3562 works on 2.4 to

3.6V, so normally it uses a 3V battery for operation. In this circuit, 9V battery voltage is reduced to the required level by R2 and ZD1. The 9V battery helps in producing a louder alarm sound from loudspeaker LS1.

Output of the UM3562 at pin 5 is in audio range. The two transistors T1 and T2 used as a Darlington pair amplify the sound.

Working of the circuit is simple.